The University of New South Wales

School of Electrical Engineering and Telecommunications

Sample Final Exam Paper 2

Candidates should attempt ALL questions

1. Time allowed: **3** hours.

2. The paper contains **5** questions.

3. The questions are NOT of equal value. Part marks are as indicated .

4. The examination paper may be retained by the candidate.

5. Casio $f\chi$ - 911W scientific calculator provided by examination section.

6. Transform tables are supplied for your use.

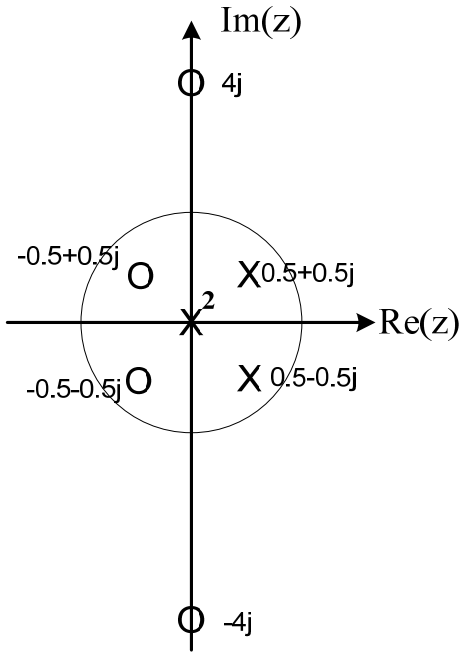
ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY ONLY BE USED FOR DRAWING, SKETCHING OR GRAPHICAL WORK.

Question 1. (30 marks)

A. The pole-zero diagram of a discrete-time system $H(z)$ is shown in Fig 12. Find expressions for a minimum-phase system $H_1(z)$ and an all pass system $H_{ap}(z)$ such that

$$H(z) = H_1(z)\, H_{ap}(z)$$

[6 marks]



B. Determine the coefficients of a linear-phase FIR digital filter given by equation (1) such that its frequency response is normalised so that H(0) = 1.

$$y[n] = a_0 x[n] + (1/3)\, x[n-1] + a_2\, x[n-2] \qquad (1)$$

[4 marks]

C. By determining the transfer functions for Figure 2, show that the systems (i) and (ii) are equivalent. **[8 marks]**

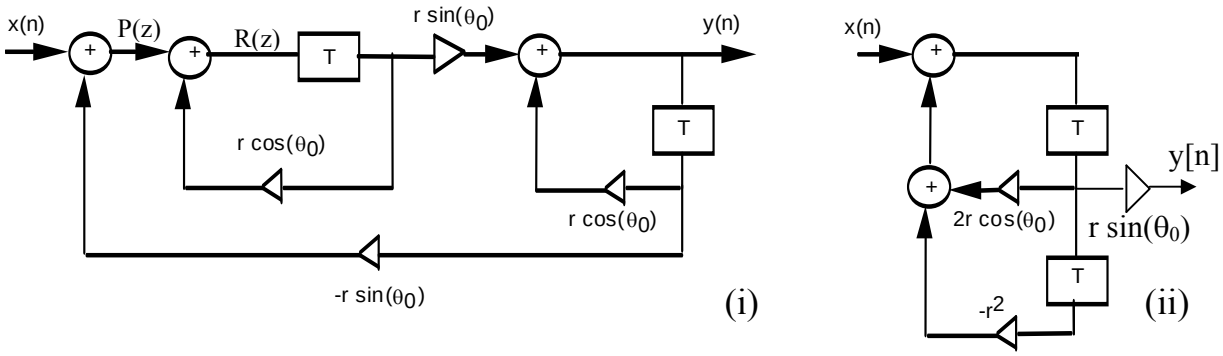


Figure 2

D. Determine and sketch the magnitude and phase responses of the system shown in Figure 4.

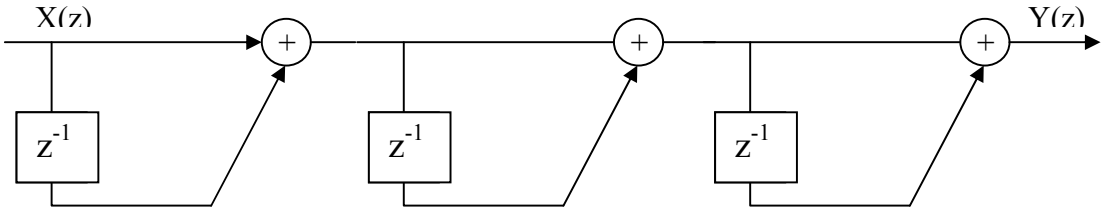


Figure 4 **[6 marks]**

E. A two-pole digital lowpass filter is given by

$$H(z) = \frac{k^2{}_0}{(1 - b_1 z^{-1})^2}$$

Determine the values of k_0 and b_1 such that the frequency response $H(\theta)$ satisfy the conditions

$$H(0) = 1; \quad \left|H(\theta)\right|^2_{\theta=\frac{\pi}{3}} = \frac{1}{4}$$

[6 marks]

Question 2. (25 marks)

A. A digital oscillator that simultaneously generates cosine and sine waves is required for telecommunications applications. Using the following equations you are required to design an oscillator with an oscillation frequency of θ .

$$\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos(n\theta)$$

$$\cos(n+1)\theta - \cos(n-1)\theta = -2\sin\theta\sin(n\theta)$$

(i) Determine the difference equations for the digital oscillator. **[5 Marks]**

(ii) Draw a structure for this oscillator with 4 multipliers, where two of the multipliers are –1. **[5 Marks]**

(iii) Determine the coefficients required to obtain the desired oscillation frequency of $3\pi/4$. **[3 Marks]**

B. (i) A FIR filter has an impulse response as shown in Figure 3.

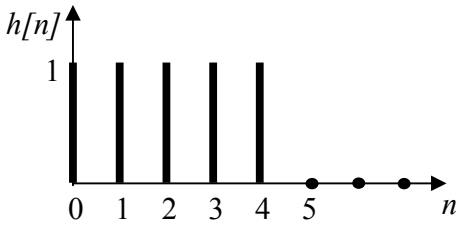


Figure 3

Show that the frequency response of this filter is given by

$$H(\theta) = e^{-j(2\theta)} \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{\theta}{2}\right)}$$ where θ is the digital frequency.

[5 marks]

(ii) Draw an implementation of this filter using pole and zero configuration.

[4 marks]

(iii) Show that the above filter has a constant group delay. **[3 marks]**

Question 3. (10 marks)

A. Evaluate the DTFT (discrete time Fourier transform) of

$$x[n] = \begin{cases} 1, & -N \leq n \leq N \\ 0, & otherwise \end{cases}$$

[5 marks]

B. Show that,

$$y[n] = \begin{cases} 2N + 1 + n, & -2N \leq n \leq 0 \\ 2N + 1 - n, & 0 \leq n \leq 2N \\ 0, & otherwise \end{cases}$$

Is given by $y[n] = x[n] * x[n]$. Hence use the DTFT of $x[n]$ to determine the DTFT of $y[n]$.

[5 marks]

Question 4. (15 marks)

A. Let a causal linear time invariant discrete-time system be characterised by a real impulse response *h[n]* with a DTFT of $H(\theta)$. Consider the system of Figure 6, where *x[n]* is a finite length sequence. Determine the frequency response of the overall system $G(\theta)$ in terms of $H(\theta)$, and show that it has a zero-phase response. **[5 marks]**

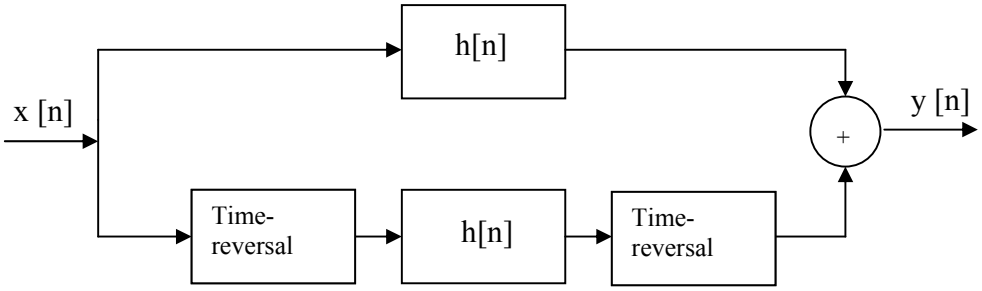


Figure 6

B. It is required to design a digital filter to approximate the analogue filter given by the following transfer function

$$H(s) = \frac{0.1s}{s^2 + 0.1s + 1}$$

(i) Using the bilinear transform show that the resulting digital filter can be written in the form of

$$H(z) = \frac{(a_0 - a_0 z^{-2})}{1 + b_2 z^{-2}}$$

Determine the values of the coefficients a_0 and b_2. The resulting digital filter is to have a resonant frequency of 600*Hz*. The sampling frequency is 2400 Hz. **[7 marks]**

(ii) Draw a canonic realisation of $H(z)$. **[3 marks]**

Question 5. (20 marks)

A. A signal has been sampled at 44 kHz and for a particular application this signal is required to be used at a reduced sampling rate of 8 kHz.

(i) Draw a block diagram to show how you would achieve this. **[4 marks]**

(ii) Indicate the type of digital filter which you would choose for part (i) above (give reasons).

. **[3 marks]**

B. The sampling period T of the input signal shown in Figure 5 is 125 μs. The relative frequency is θ = ωT. The first oscillator generates a carrier with a relative frequency $\theta_1 = \omega_1 T$, where $\omega_1 = 2\pi.2.10^3$ *rad* / sec. The second oscillator generates a carrier with a relative frequency $\theta_2 = (\omega_c + \omega_1)\, T$, where $\omega_c = 2\pi .10^3$ *rad* / sec. The low-pass filter has the following characteristics:

$$G(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \dfrac{\pi}{4} \\ 0 & \dfrac{\pi}{4} < |\theta| \le \pi \end{cases}$$

If the input spectrum is $X(\theta) = 1 + \cos(\theta) \quad -\pi \le \theta \le \pi$,

(i) Write appropriate equations for $B_c(\theta)$, $D_c(\theta)$, and $R_c(\theta)$.

[6 Marks]

(ii) Sketch |X(θ)|, $|B_c(\theta)|$, $|D_c(\theta)|$, and $|R_c(\theta)|$ against θ,

[7 Marks]

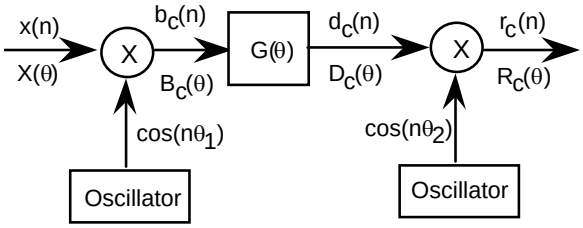


Figure 5

Note: $x(n)\cos(\theta_0 n) \quad \Leftrightarrow \quad \frac{1}{2}\left[X(\theta - \theta_0) + X(\theta + \theta_0)\right]$

Table of the z transforms of useful sequences

Basic z-Transforms	**Sequence**	**z-Transform**
1. Unit sample	$\delta[n]$	1
	$\delta[n-k]$	z^{-k}
2. Unit step	u[n]	z/(z-1)
3. Exponential	a^n u[n)]	z/(z-a)
4. Sinusoidal	$\sin[\theta_o n]$ u[n]	$\frac{z \sin\theta_0}{z^2 - 2\cos\theta_0\ z + 1}$
	$\cos[\theta_o n]$ u[n]	$\frac{z^2 - z\cos\theta_0}{z^2 - 2\cos\theta_0\ z + 1}$
5. Unit ramp	nu[n]	$\frac{z}{(z-1)^2}$

$$1 + z^{-1} + z^{-2} + z^{-3} + \ldots\ldots\ldots + z^{-(N-1)} = \frac{1 - z^{-N}}{1 - z^{-1}}$$

z-Transform Properties	**Sequence**	**z-Transform**
	x[n]	X(z)
	x[n-k]	z^{-k} X(z)
	a^n x[n]	X(z/a)
	x[-n]	$X(z^{-1})$
	x[n]*y[n]	X(z).Y(z)
	n x[n]	-z (dx(z)/dz)

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\cos\theta = \frac{1}{2}\left[e^{j\theta} + e^{-j\theta}\right]$$
$$\sin\theta = \frac{1}{2j}\left[e^{j\theta} - e^{-j\theta}\right]$$

Fourier Series:

$$x(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t); \qquad \omega_0 = 2\pi f_0; \quad f_0 = \frac{1}{T}$$

The fundamental frequency of x(t) is ω_0 and T is the period of the signal x(t).

$$A_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\cos(n\omega_0 t)\,dt \qquad n = 0,1,2.... \qquad B_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\sin(n\omega_0 t)\,dt \qquad n = 1,2,3...$$

Fourier Transform pair (for discrete-time signal)

$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\theta}... \qquad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\theta)e^{jn\theta}\,d\theta$$

$\theta = digital\ frequency; \quad -\pi \le \theta \le \pi$

A **Butterworth filter** of order'n' is defined by

$$|H(j\omega)|^2 = \frac{1}{1+\left(\frac{\omega}{\omega_C}\right)^{2n}}$$ where ω_c is the cut-off frequency of the filter

n	Butterworth Polynomials
1	$S + 1$
2	$s^2 + \sqrt{2}s + 1$
3	$(s^2 + s + 1)(s + 1)$

Impulse Invariant Method:

$$\frac{1}{s+b} \rightarrow \frac{1}{1 - e^{-bT}z^{-1}} \qquad b > 0$$ *T*-sampling period

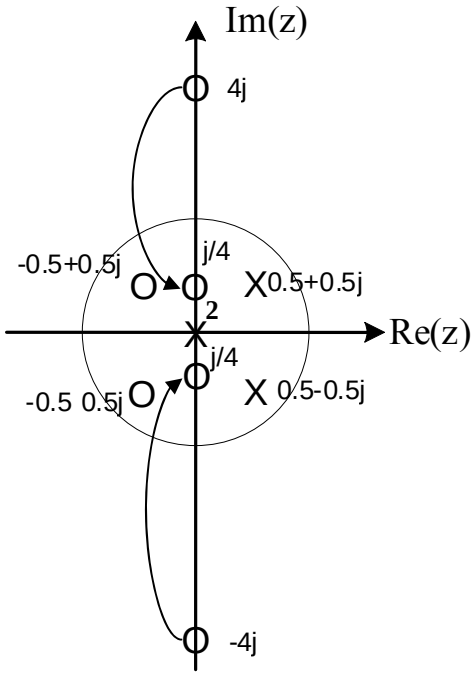
Bilinear Transformation:

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}} \qquad T - sampling \quad period$$

End of paper

Sample Final Exam Paper 2
Model Answers

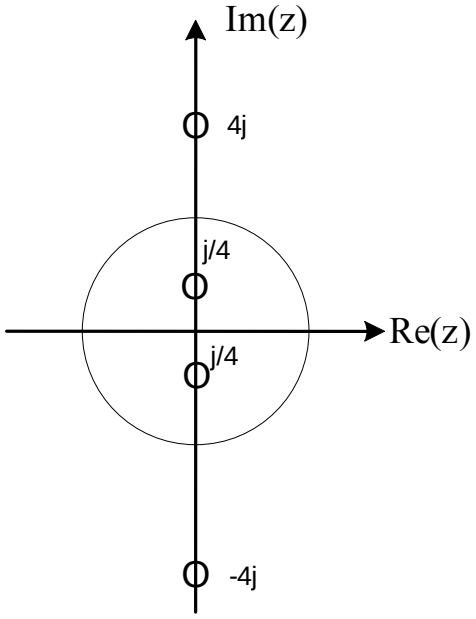
Question 1
(A)
Minimum phase



$$H_l(z) = \frac{(z - j/4)(z + j/4)(z + 1/2 - j/2)(z + 1/2 + j/2)}{(z - 1/2 - j/2)(z - 1/2 + j/2)z^2}$$

$$= \frac{(z - j/16)\left((z + 1/2)^2 + 1/4\right)}{\left((z - 1/2)^2 + 1/4\right)z^2}$$

All pass



$$H_{ap}(z)=\frac{(z-4)(z+4)}{(z-j/4)(z+j/4)}$$
$$\therefore H(z)=H_l(z)H_{ap}(z)$$

(B)

(i)

$$H(z)=a_0+\frac{1}{3}z^{-1}+a_2z^{-2}$$

$$H(\theta)\big|_{\theta=0}=1=a_0+\frac{1}{3}+a_2$$

For linear phase $a_0=a_2$

$$1=a_0+\frac{1}{3}+a_0$$

$$\therefore a_0=\frac{1}{3},\quad a_2=\frac{1}{3}$$

(C)

(i)

$$R(z)=X(z)+z^{-1}Y(z)(-r\sin\theta_0)+r\cos\theta_0R(z)z^{-1}$$
$$R(z)(1-r\cos\theta_0z^{-1})=X(z)-r\sin\theta_0z^{-1}Y(z)\quad ----(1)$$
$$Y(z)=R(z)r\sin\theta_0z^{-1}+Y(z)r\cos\theta_0z^{-1}$$
$$R(z)=\frac{Y(z)[1-r\cos\theta_0z^{-1}]}{r\sin\theta_0z^{-1}}\quad -----(2)$$

From (1) and (2):

$$\frac{X(z)-r\sin\theta_0z^{-1}Y(z)}{1-r\cos\theta_0z^{-1}}=\frac{Y(z)[1-r\cos\theta_0z^{-1}]}{r\sin\theta_0z^{-1}}$$
$$X(z)r\sin\theta_0z^{-1}-r^2\sin^2\theta_0z^{-2}Y(z)=Y(z)[1-2r\cos\theta_0z^{-1}+r^2\cos^2\theta_0z^{-2}]$$
$$\therefore\frac{Y(z)}{X(z)}=\frac{r\sin\theta_0z^{-1}}{1-2r\cos\theta_0z^{-1}+r^2\theta_0z^{-2}}$$

(ii)

$$\frac{Y(z)}{X(z)} = \frac{r\sin\theta_0 z^{-1}}{1 - 2r\cos\theta_0 z^{-1} + r^2\theta_0 z^{-2}}$$

(D)

$$H(z) = \left(1 - z^{-1}\right)^3$$

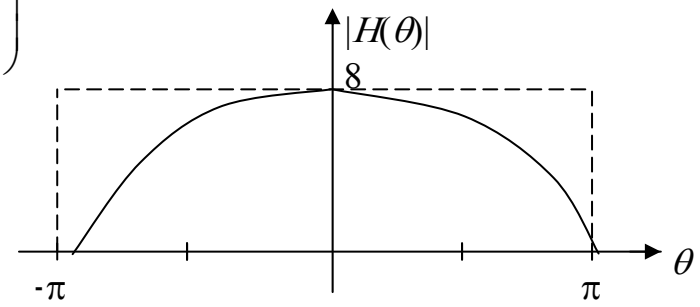
$$H(\theta) = \left(1 - e^{-j\theta}\right)^3$$

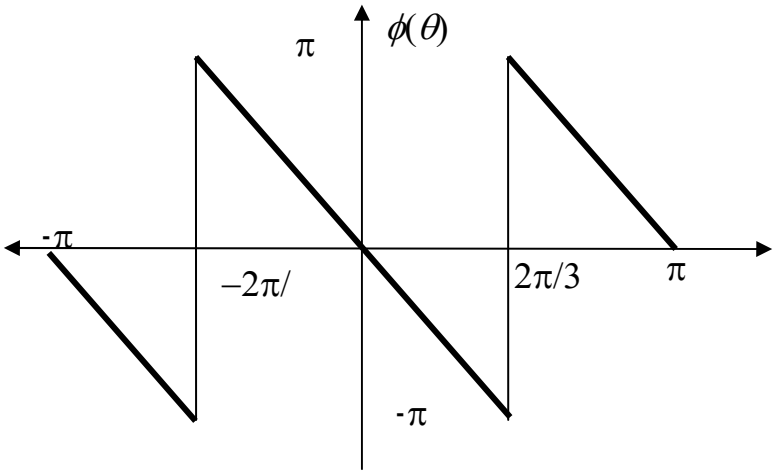
$$= \left(-e^{-j\frac{\theta}{2}}\right)^3 \left(e^{j\frac{\theta}{2}} + e^{-j\frac{\theta}{2}}\right)^3$$

$$= e^{-j\frac{3\theta}{2}} \times 8 \times \cos^3\frac{\theta}{2}$$

$$\therefore |H(\theta)| = \left|8\cos^3\frac{\theta}{2}\right|$$

$$\therefore \phi(\theta) = -\tfrac{3}{2}\theta$$





(E)

$$H(\theta) = \frac{k_0^2}{\left(1 - b_1 e^{-j\theta}\right)^2}$$

$$H(\theta)\big|_{\theta=0} = 1 \quad \Rightarrow \quad 1 = \frac{k_0^2}{(1-b_1)^2} \quad \Rightarrow k_0^2 = (1-b_1)^2$$

$$|H(\theta)|^2 = \frac{k_0^4}{\left(1 + b_1^2 - 2b\cos\theta\right)^2}$$

$$|H(\theta)|^2_{\theta=\frac{\pi}{4}} = \tfrac{1}{4} = \frac{k_0^4}{\left(1 + b_1^2 - 2b\times\frac{1}{2}\right)^2}$$

$$4(1-b_1)^4 = (1 + b_1^2 - b_1)^2$$

Taking square root on both side:

$2(1-b_1)^2 = 1+b_1^2 - b_1 \quad - \quad eqn(1)$

$2(1-b_1)^2 = -(1+b_1^2 - b_1) \quad - \quad eqn(2)$

From eqn(1):

$b_1 = 0.38 \quad or \quad b_1 = 2.62$

but $b_1 = 2.62$ is >1 → unstable system

$\therefore b_1 = 0.38$

$\therefore k_0 = 1 - b_1 = 0.6180$

From eqn(2): solving the quadratic equation results in an imaginary part in the solution so it's invalid.

Question 2

(A)

(i)

From the 1st equation:

$$\cos(n+1)\theta = 2\cos\theta + \cos n\theta - \cos(n-1)\theta$$

From the 2nd equation:

$$\sin n\theta = \frac{\cos(n+1)\theta - \cos(n-1)\theta}{-2\sin\theta}$$

Let $y_1(n) = \cos n\theta$

$y_2(n) = \sin n\theta$

$$y_1(n+1) = 2\cos\theta y_1(n) - y_1(n-1) \quad ----(1)$$

$$y_2(n) = \frac{1}{2\sin\theta}\left[y_1(n-1) - y_1(n+1)\right] \quad ----(2)$$

(ii) Draw the block diagram from the above calculation.

(iii)

$\theta = \frac{3}{4}\pi$

$\therefore a = 2\cos\frac{3}{4}\pi = -= \sqrt{2}$

$$b = \frac{1}{2\sin\frac{3}{4}\pi} = \frac{\sqrt{2}}{2}$$

B

(i)

$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

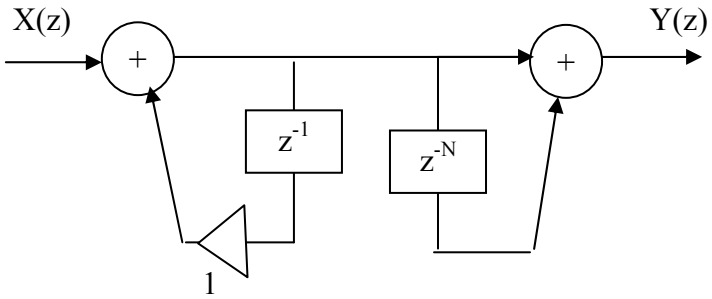
$$H(z) = \frac{1 - z^{-5}}{1 - z^{-1}}$$

$$H(\theta) = \frac{1 - e^{-j5\theta}}{1 - e^{-j\theta}} = \frac{e^{-j\frac{5}{2}\theta}}{e^{-j\frac{1}{2}\theta}} \frac{\left(e^{j\frac{5}{2}\theta} - e^{-j\frac{5}{2}\theta}\right)}{\left(e^{-j\frac{1}{2}\theta} - e^{-j\frac{1}{2}\theta}\right)}$$

$$H(\theta) = e^{-j2\theta} \frac{\sin\frac{5}{2}\theta}{\sin\frac{1}{2}\theta}$$

(ii)

$$H(z) = \frac{1 - z^{-5}}{1 - z^{-1}}$$



(iii)

$$\phi(\theta) = -2\theta$$

$$\therefore \tau_g = -\frac{d\phi(\theta)}{d\theta} = 2 \quad \text{(constant group delay)}$$

Question 3:

(A)

$$x[n]=\begin{cases}1 & -N\le n\le N\\ 0 & otherwise\end{cases}$$

$$Let \quad X(\theta)\xleftrightarrow{DTFT}x[n]$$

$$X(\theta)=\sum_{n=-\infty}^{\infty}x[n]e^{-j\theta n}=\sum_{n=-N}^{N}e^{-j\theta n}$$

$$Let \quad m=n+N$$

$$X(\theta)=\sum_{m=0}^{2N}e^{-j\theta(m-N)}=e^{j\theta N}\sum_{m=0}^{2N}e^{-j\theta m}$$

$$=e^{j\theta N}\left[\frac{1-e^{-j\theta(2N+1)}}{1-e^{-j\theta}}\right]$$

$$=\frac{e^{j\theta N}-e^{-j\theta N}e^{-j\theta}}{1-e^{-j\theta}}$$

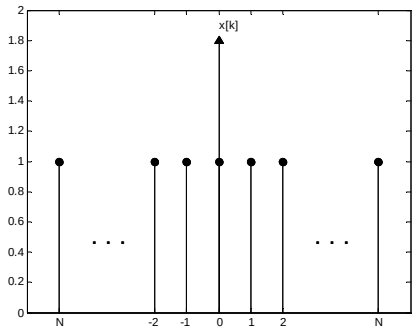
$$=\frac{e^{j\theta N}e^{j\frac{\theta}{2}}-e^{-j\theta N}e^{-j\frac{\theta}{2}}}{e^{j\frac{\theta}{2}}-e^{-j\frac{\theta}{2}}}$$

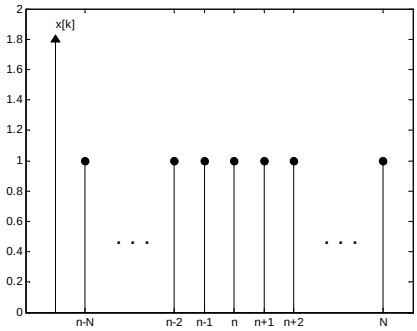
$$=\frac{e^{j\theta\left(N+\frac{1}{2}\right)}-e^{-j\theta\left(N+\frac{1}{2}\right)}}{e^{j\frac{\theta}{2}}-e^{-j\frac{\theta}{2}}}$$

$$=\frac{\sin\left[\left(N+\frac{1}{2}\right)\theta\right]}{\sin\frac{\theta}{2}}=\frac{\sin\left[(2N+1)\frac{\theta}{2}\right]}{\sin\frac{\theta}{2}}$$

(B)

$$x[n]*x[n]=\sum_{k=-\infty}^{\infty}x(k)x(n-k)$$





$For\ n < -2N \quad x(k)x(n-k) = 0 \quad \forall k$

$For\ -2N \le n \le 0 \quad x(k)x(n-k) = \begin{cases} 1 & -N < k \le n+N \\ 0 & otherwise \end{cases}$

$For\ 0 \le n \le 2N \quad x(k)x(n-k) = \begin{cases} 1 & n-N < k \le N \\ 0 & otherwise \end{cases}$

$For\ n > 2N \quad x(k)x(n-k) = 0 \quad \forall k$

$\therefore x[n] * x[n]$

$$= \begin{cases} \sum_{k=-N}^{n+N} 1 & -2N \le n \le 0 \\ \sum_{k=n-N}^{N} 1 & 0 \le n \le 2N \end{cases}$$

$$= \begin{cases} 2N+n & -2N \le n \le 0 \\ 2N-n & 0 \le n \le 2N \end{cases}$$

$= y[n]$

$Let \quad Y(\theta) \xleftrightarrow{DTFT} y[n]$

$Y(\theta) \xleftrightarrow{DTFT} x[n] * x[n]$

$\therefore Y(\theta) = X(\theta)X(\theta) \quad (\text{Convolution property})$

Question 4:

(A)

$h(n) \xrightarrow{DTFT} H(\theta)$

$v(n) = x(-n) * h(n)$

$u(n) = v(-n) = x(n) * h(n)$

$y(n) = \{h(n) + h(-n)\} * x(n)$

$Y(z) = \{H(z) + H(z^{-1})\} \bullet X(z)$

$G(z) = H(z) + H(z^{-1})$

$G(\theta) = H(\theta) + H^*(\theta) = (a + jb) + (a - jb) = 2a$

Hence the equivalent transfer function $G(\theta)$ is real and has zero phase.

(B)

$$H(s)=\frac{0.1s}{s^2+0.1s+1}$$

$$\theta_C=2\pi\frac{fc}{fs}=\frac{\pi}{2}$$

$$\therefore \omega'_C=\frac{2}{T}\tan\left(\frac{\theta_C}{2}\right)=\frac{2}{T}\ \text{rad/sec}$$

Pre-warped analogue filter:

$$H(s)=\frac{0.1\times\frac{s}{\frac{2}{T}}}{\left(\frac{s}{\frac{2}{T}}\right)^2+0.1\times\frac{s}{\frac{2}{T}}+1}$$

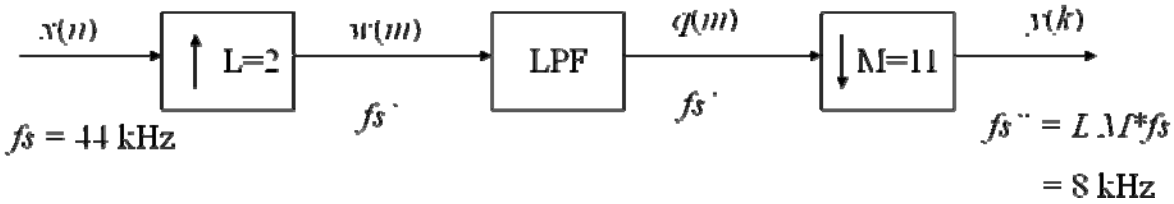
$$H(s)\big|_{s=\frac{2}{T}\frac{z-1}{z+1}}=H(z)=\frac{0.1\times\frac{2}{T}\times\frac{2}{T}\frac{z-1}{z+1}}{\left(\frac{2}{T}\right)^2\left(\frac{z-1}{z+1}\right)^2+0.1\times\frac{2}{T}\times\frac{2}{T}\frac{z-1}{z+1}+\left(\frac{2}{T}\right)^2}$$

$$H(z)=\frac{1}{21}\frac{z^2-1}{z^2+\frac{19}{21}}=\frac{1}{21}\frac{1-z^{-2}}{1+\frac{19}{21}z^{-2}}$$

Question 5:

(A)

(i)



$$\frac{L}{M}=\frac{8000}{44000}=\frac{2}{11}$$

(ii)

Using IIR filter, we cannot take advantage of the fact that we have to compute only every *N* th output, since previous outputs are needed to compute the *N* th output.

On the other hand, using FIR filter we can do our computation at the rate of fs'/N → a significantly lower computational rate.

(B)

$$\theta_1 = 2\pi \times 2 \times 10^3 \times \frac{1}{8\times 10^3} = \frac{\pi}{2}$$

$$\theta_2 = \omega_C T + \omega_1 T = \frac{\pi}{2} + 2\pi \times 10^3 \times \frac{1}{8\times 10^3} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

$$X(\theta) = 1 + \cos\theta$$

$$B_C(\theta) = \frac{1}{2}\left[X(\theta-\theta_1) + X(\theta+\theta_1)\right] = \frac{1}{2}\left[1 + \cos\left(\theta - \frac{\pi}{2}\right) + 1 + \cos\left(\theta + \frac{\pi}{2}\right)\right] = \frac{1}{2}\times 2 = 1$$

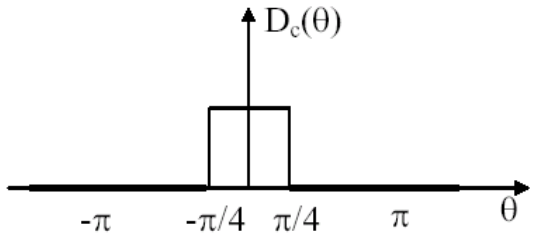
$$D_C(\theta) = B_C(\theta)\cdot G(\theta) = 1, \quad 0 < |\theta| < \tfrac{\pi}{4}$$

$$R_C(\theta) = \frac{1}{2}\left[D_C(\theta-\theta_2) + D_C(\theta+\theta_2)\right] = \frac{1}{2}\left[D_C\left(\theta - \tfrac{3}{4}\pi\right) + D_C\left(\theta + \tfrac{3}{4}\pi\right)\right]$$

$$R_C(\theta) = \begin{cases} \frac{1}{2} & -\frac{\pi}{4} \le \theta - \frac{3\pi}{4} \le \frac{\pi}{4} \Rightarrow \frac{\pi}{2} \le \theta \le \pi \\ \frac{1}{2} & -\frac{\pi}{4} \le \theta + \frac{3\pi}{4} \le \frac{\pi}{4} \Rightarrow -\pi \le \theta \le -\frac{\pi}{2} \\ 0 & \text{elsewhere} \end{cases}$$

Then from the above calculation, sketch $X(\theta), B_C(\theta), D_C(\theta)$, and $R_C(\theta)$.

$$\mathrm{B_c}(\theta) = 1, \mathrm{D_c}(\theta) = 1$$

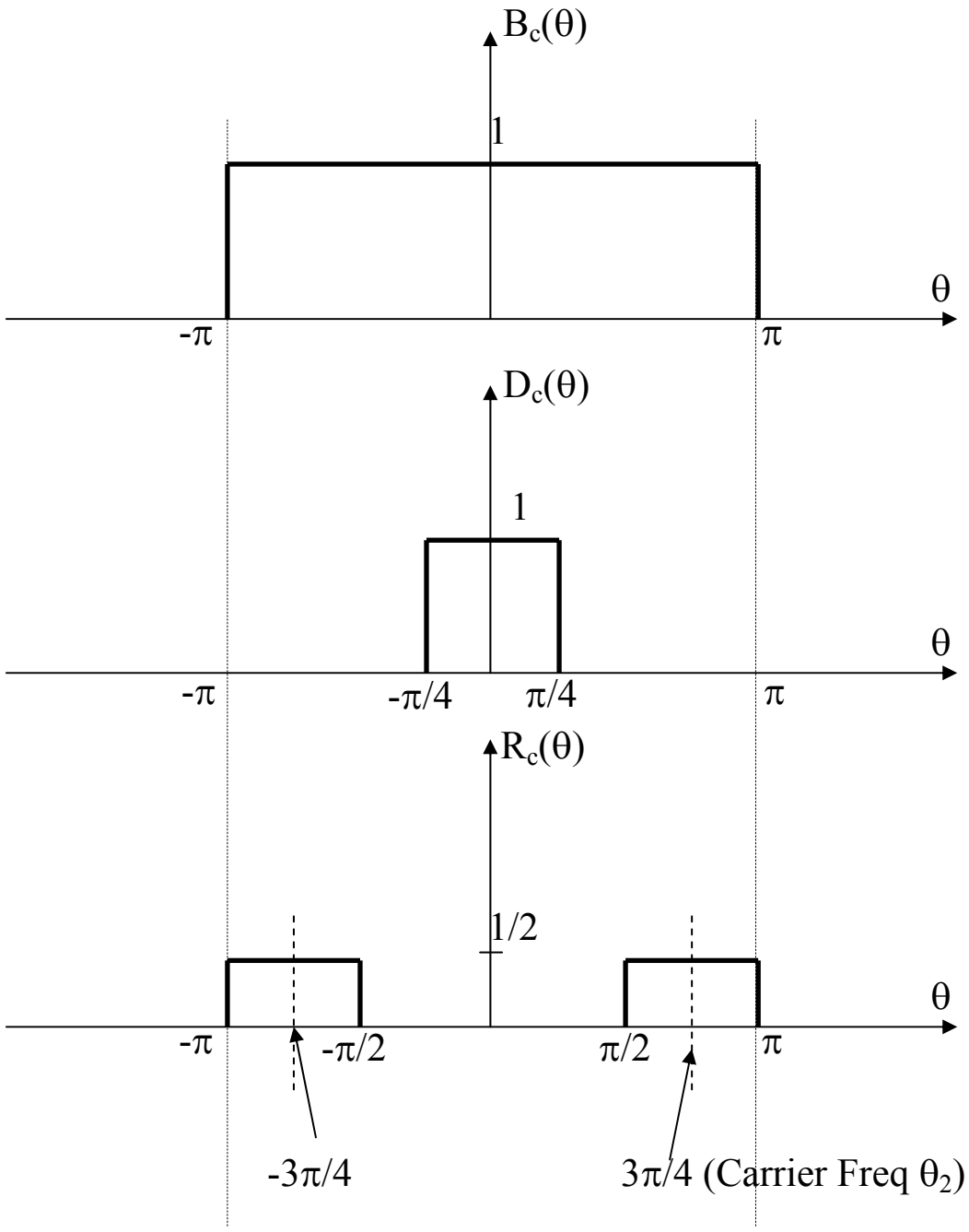


$$\mathrm{R_c}(\theta) = \frac{1}{2}\left\{\mathrm{D_c}(\theta - \theta_1) + \mathrm{D_c}(\theta + \theta_1)\right\}$$

$$= \frac{1}{2}\left\{ D_C(\theta - \frac{3\pi}{4}) + D_c(\theta + \frac{3\pi}{4}) \right\}$$

$$= \begin{cases} \frac{1}{2} & \frac{-\pi}{4}(\theta - \frac{3\pi}{4}) \le \frac{\pi}{4} \\ \\ \frac{1}{2} & \frac{-\pi}{4} \le \theta + \frac{3\pi}{4} \le \frac{\pi}{4} \\ \\ 0 & \text{elsewhere} \end{cases}$$

$$= \begin{cases} \frac{1}{2} & \frac{\pi}{2} \le \theta \le \pi \\ \\ \frac{1}{2} & -\pi \le \theta \le \frac{-\pi}{2} \\ \\ 0 & \text{elsewhere} \end{cases}$$


$B_c(\theta)$
1
θ
-π
π
$D_c(\theta)$
1
θ
-π
-π/4
π/4
π
$R_c(\theta)$
1/2
θ
-π
-π/2
π/2
π
-3π/4
3π/4 (Carrier Freq θ_2)